

DRUGS USED IN RHEUMATOID ATHRITIS

BIETE LUNDAU LUKE

DipPharm, BPharm, MclinPharm

Introduction

- ▶ Rheumatoid arthritis (RA) is an autoimmune disorder of unknown cause
- ▶ RA is characterized with symmetrical joint inflammation frequently affects the small joints of the hands, wrists, and also the joints of the ankles, elbows, hips, knees and shoulders
- ▶ Most patients experiencing a chronic, fluctuating course of disease that despite the therapeutic measures results in progressive joint destruction, deformity, disability and premature death
- ▶ It is characterized by chronic inflammation in the synovial membrane of affected joints that ultimately leads to loss of daily function due to chronic pain and fatigue
- ▶ The majority of patients also have deterioration of cartilage and bone in the affected joints, which may eventually lead to permanent disability
- ▶ Rheumatoid arthritis is associated with increased morbidity , It affecting approximately 1 % of the adult population

Onset of RA

- ▶ Although Rheumatoid arthritis may present at any age, patients most commonly are first affected in the third to sixth decades
- ▶ Female: male 3:1
- ▶ Initial pattern of joint involvement could be:-
 - 1) Polyarticular : most common
 - 2) Oligoarticular
 - 3) Monoarticular

Morning joint stiffness > 1 hour and easing with physical activity is characteristic. Small joints of hand and feet are typically involved

Extra articular manifestations

- ▶ **Heart** - pericarditis , cardiomyopathy and valvular incompetence
- ▶ **Lungs** - rheumatoid lung (honeycombed appearance on CXR due to multiple nodules, caplan syndrome (associated with pneumoconiosis), idiopathic pulmonary fibrosis, intestinal fibrosis
- ▶ **Eyes** - scleritis(most common ocular complication of RA, scleromalacia perforans
- ▶ **Hematopoietic system** –Felty's syndrome (anemia, splenomegaly and leukopenia

Goals of management

Early aggressive individualized treatment is recommended which should aim at the following;

- ▶ Relieving pain
- ▶ Preventing damage or disability
- ▶ Patient education about the disease
- ▶ Physical therapy for stretching and other range of motion exercises
- ▶ Occupational therapy for splints and adaptive devices
- ▶ Treatment should be started early and should be individualized

Treatment

NSAIDs

- ▶ Non-Steroidal anti-inflammatory drugs (NSAIDs) used for the control of symptoms of pain and swelling by inhibiting Cyclo oxygenase Enzyme (COX)
- ▶ They however, have the following limitations;
 - 1) Do not alter course of the disease
 - 2) Chronic use should be minimized to avoid unnecessary harm from side effects
 - 3) Most common side effect related to GI tract

Disease modifying anti-rheumatic agents (DMARDs)

- ▶ Drugs that actually alter the disease course
- ▶ Should be used as soon as diagnosis is made
- ▶ Appearance of benefit delayed for weeks to months
- ▶ NSAIDS must be continued with them until true remission is achieved
- ▶ Induction of true remission is unusual

DMARDS

Most commonly used

- ▶ Methotrexate
- ▶ Leflunomide
- ▶ Sulfasalazine
- ▶ Hydroxychloroquine

Less commonly used

- ▶ Chloroquine
- ▶ Gold
- ▶ Cyclosporine A
- ▶ D-penicillamine
- ▶ Azathioprine
- ▶ Cyclophosphamide

Specific Drugs

Methotrexate

- ▶ An antineoplastic and immunomodulating drug which remains the single most effective DMARD and is the drug of choice in most RA patients

Mechanism of action

- ▶ Inhibits human folate reductase thereby reducing the available active form of folate required for thymidylate and DNA synthesis
- ▶ Inhibits lymphocyte proliferation and the production of cytokines and rheumatoid factor
- ▶ Interferes with polymorphonuclear leukocyte chemotaxis and reduces the production of cytokines and free radicals that damage the synovial membrane and bone

Side effects of methotrexate

Treatment with methotrexate is generally well tolerated by patients with RA but it is reported the following side effects;

- ▶ GIT upset
- ▶ Hepatotoxicity
- ▶ Bone marrow suppression
- ▶ Pulmonary fibrosis

Hydroxychloroquine

- ▶ An antimalarial related to chloroquine is extensively used as a DMARD

Mechanism of action

- ▶ Reduces the chemotaxis and phagocytosis of polymorphonuclear leukocytes
- ▶ Decreases the production of superoxide radicals
- ▶ Has slow onset of action and can take up to 6 months before benefits are observed

Side effects

- ▶ Occasionally causes GIT disturbances
- ▶ Patients receiving hydroxychloroquine should be monitored for ocular adverse effects including blurred vision, scotomas, night blindness

Advantage of Hydroxychloroquine over other DMARDs

Unlike many other DMARDs, Hydroxychloroquine does not cause the following;

- ▶ Myelosuppression
- ▶ Hepatic toxicity
- ▶ Renal toxicity

Gold salts

- ▶ Examples; Auranofin, Gold sodium thiomalate (sodium aurothiomalate)
- ▶ Once used extensively in the treatment of RA but has lost its popularity because of the introduction of newer, more efficacious and safer newer DMARDs
- ▶ Anti-rheumatic effects are usually not seen until about 3 to 6 months after starting therapy

Mechanism of action

- ▶ Not well understood but is suggested to be linked to anti-mitochondrial activity and cell apoptosis

Side effects of Gold salts

- ▶ Hematologic effects
- ▶ Dermatologic effects
- ▶ GIT effects
- ▶ Flushing
- ▶ Hypotension and tachycardia
- ▶ Skin and stomatitis are commonly observed and require discontinuation of treatment until they resolve

When to start DMARDs

- ▶ DMARDs are indicated in all patients with RA who continue to have active disease even after 3 months of NSAIDS use
- ▶ The period of 3 months is arbitrary & has been chosen since a small percentage of patients may go in spontaneous remission
- ▶ The vast majority , however , need DMARDs and many rheumatologists start DMARDs from Day 1

Glucocorticoids

- ▶ Example – prednisolone, methylprednisolone
- ▶ Has played an important role in the treatment of RA

Mechanism of action

- ▶ Induces the formation of lipocortin, a protein that inhibits phospholipase A₂ activity
- ▶ This leads to arachidonic acid inhibition and the subsequent inhibition of prostaglandin formation
- ▶ Glucocorticoids also inhibit the production of cytokines, including interleukins and tumor necrosis factor (TNF)

Corticosteroids

- ▶ Corticosteroids, both systemic and intra-articular are important adjuncts in management of RA

Indications for systemic steroids are:

- ▶ For treatment of rheumatoid flares
- ▶ For extra-articular RA like rheumatoid vasculitis and interstitial lung disease
- ▶ As bridge therapy for 6-8 weeks before the action of DMARDs begin
- ▶ Maintenance dose of 10mg or less of prednisolone daily in patients with active RA
- ▶ Sometimes in pregnancy when other DMARDs cannot be used

Indication of glucocorticoids

- ▶ Glucocorticoids act more rapidly than other DMARDs but their long term use is limited by the development of adverse effects
- ▶ They are therefore, used in various ways to manage patients with RA and other inflammatory joint diseases
- ▶ Used to induce remission of the disease at the time when other slow DMARD is started to provide short courses of therapy during disease flare ups
- ▶ Also used to provide continuous low doses background therapy in patients being treated with other DMARDs and NSAIDs
shot courses of therapy

Selection of DMARDs

- ▶ There are no strict guidelines about which DMARDs to start first in an individual
- ▶ Methotrexate has rapid onset of action than other DMARD
- ▶ Taking in account patient tolerance, cost considerations and ease of once weekly oral administration METHOTREXATE is the DMARD of choice, most widely prescribed in the world

DMARDs - alone or in combination

- ▶ Since single DMARD therapy (in conjunction with NSAIDs) is often only modestly effective , combination therapy has an inherent approach
- ▶ DMARD combination is specially effective if they include methotrexate as an anchor drug
- ▶ Combination of methotrexate with leflunamide are synergistic since there mode of action is different

Limitations of conventional DMARDs

- ▶ The onset of action takes several months
- ▶ The remission induced in many cases is partial
- ▶ There may be substantial toxicity which requires careful monitoring
- ▶ DMARDs have a tendency to lose effectiveness with time-(slip out)

Biological agents

- ▶ These drawbacks noted with DMARDS made researchers look for alternative treatment strategies for RA and these are Biologic Response Modifiers
- ▶ Biologic agents are produced in a living system like a microorganism, plant or an animal cell by harnessing rDNA
- ▶ The end product is a protein directed against a specific gene or another protein which is highly specific for its function and target antigen
- ▶ The introduction of biological such as erythropoietin, insulin, growth hormones and anti-cytokine therapies brought a new era in modern medicine and transformed the treatment of many chronic diseases

Immuno-modulators

- ▶ Examples of these biological agents are Immuno-modulators e.g. etanercept, infliximab, adalimumab, certolimumab

Ertanercept

- ▶ Protein formed by recombining human soluble TNF α receptor with fragments of human immunoglobulin G1 (IgG1)
- ▶ Antagonises the TNF α to a greater extent and has a long half life
- ▶ Intended for use in patients whose RA is refractory to treatment with methotrexate or other DMARDs

Infliximab

- ▶ Is chimeric IgG1 monoclonal antibody that inactivates TNF α
- ▶ Found to be more effective when combined with methotrexate than when used alone

Adalimumab

- ▶ Is a human TNF α made by recombinant DNA technology
- ▶ Methotrexate, glucocorticoids, salicylates , NSAIDs analgesics and other DMARDs can be used safely

END